Why Structured Semantic Streams Improve LLM Reasoning over Long Documents

A technical note for engineers and researchers

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How to read this paper. This paper has two layers. Sections marked **THEORY** are theoretical background. They explain why the observed effects occur at a deeper level. They can be skipped without any loss of understanding of the technical argument. Readers who want only the engineering picture can proceed directly from Section 1 to Section 3.

1. The Problem: Why Long Documents Break LLMs

Large language models process text as token sequences and use self-attention to integrate information across the context window. The nominal context window of modern models is large — in some cases hundreds of thousands of tokens. In practice, however, analytical accuracy degrades well before that limit is reached.

This degradation is not a bug or an implementation failure. It is a structural consequence of how transformers process unstructured natural language. Four mechanisms are responsible.

1.1 Attention Diffusion

Raw narrative text is lexically diverse and structurally heterogeneous. A single document may contain events, background commentary, citations, meta-discourse, and direct speech, with no explicit boundary markers between them. Attention heads must distribute their capacity across all of this material simultaneously.

As document length increases, attention weight disperses across a larger token population. Heads that successfully track a dependency at short range lose coherence as the distance between tokens grows. The result is progressive degradation of entity tracking and causal reasoning — not because the context window is exceeded, but because the useful signal is diluted by irrelevant material.

1.2 Lexical Variability and Entity Drift

In natural language, the same entity may appear under many surface forms across a document:

```
Central Intelligence Agency
CIA
```

```
the Agency
the organization
it
```

Name Mover heads — attention heads that propagate entity identity across the sequence — must resolve these forms dynamically during inference. Each resolution consumes attention capacity. Over long documents with many entities and many surface variants, this overhead accumulates. Entities lose their stable identity. The model begins substituting related but incorrect entities — a failure mode researchers call entity drift.

1.3 Implicit Event Structure

Events in natural language are rarely encoded in a consistent format. Compare:

```
The CIA retained the shellfish toxin.
The toxin remained in storage under CIA control.
Storage continued despite the Presidential directive.
```

These three sentences describe the same event state, but the actor, action, and object are encoded differently in each. Predicate-Argument attention heads — which approximate semantic role labeling — must reconstruct actor-action-object structure from variable syntax on every inference pass. This reconstruction is redundant work that consumes context capacity without contributing to analytical output.

1.4 Effective Context Collapse

The combined effect of attention diffusion, entity drift, and implicit structure is that the practical reasoning range of a model over a long document is substantially shorter than its nominal context window. Typical failure signatures include:

- Entity confusion: attributes of one actor transferred to another
- Causal chain loss: intermediate events dropped from multi-step reasoning
- Temporal disorder: events reported out of sequence
- Hallucinated bridging: the model generates synthetic connective material to fill gaps it can no longer recover from context

These failures are not random. They follow predictably from the structural properties described above.

> **THEORY** *The following section explains why these limitations are structural properties of computation over natural language, not incidental artifacts of current model scale. Engineers who are satisfied with the empirical description above can skip to Section 3.*

2. Theoretical Background: Semantic Drift as a Structural Property

> **THEORY** *This section is optional. It provides the theoretical foundation for the observations in Section 1. It can be skipped without loss of continuity.*

2.1 The Normalization Problem in Symbolic Systems

The classical approach to machine reasoning over documents requires initial normalization: converting native knowledge into a form compatible with first-order predicate calculus. Only after this conversion can inference rules be applied.

This normalization is neither trivial nor neutral. Real-world narratives encode dependencies that are fundamentally heterogeneous: dynamic causal interactions, static structural relations, temporal constraints, topological conditions, normative commitments, and domain-specific transformation processes. These dependencies are not naturally reducible to logical predicates without discarding essential structural information.

Upper ontologies have not solved this problem. Domain ontologies are irregular, internally inconsistent, and domain-specific. Cyc attempted exhaustive coverage and produced an explosion of special-case predicates. The normalization step itself becomes an irreversible commitment that forecloses the very analytical questions the system is supposed to answer.

This failure is not temporary. It is fundamental to the architecture of symbolic inference. Logical operations require consistency, non-contradiction, discrete truth values, and stable identity over time. These conditions rarely hold for knowledge extracted from open, adversarial, or naturally occurring sources — precisely the sources that intelligence analysis, legal reasoning, and operational planning depend on.

2.2 Semantic Drift Is the Same Problem in Both Paradigms

The semantic loss that occurs when narrative knowledge is converted into logical form is structurally identical to the drift observed in large language models during extended reasoning over raw text. In both cases, meaning degrades because the representational substrate is not designed to preserve the structural properties of the original knowledge.

In symbolic systems, this drift is invisible: it occurs at the formalization stage, which is treated as pre-computation and excluded from the reasoning pipeline. Semantic distinctions discarded before inference simply do not participate in subsequent computation.

In transformer models, the drift is observable: it manifests as entity confusion, causal chain loss, and hallucination. The mechanism differs — vector-based rather than symbolic — but the underlying cause is the same. The model must interleave semantic reconstruction with reasoning within a single process. This architectural entanglement is inherently unstable. Meaning must be continuously reassembled from tokens rather than read from an explicit representation.

2.3 Emergent Behavior and the Limits of Mathematical Prediction

Transformer architectures are defined by precise mathematical operations: linear projections, softmax attention, residual connections, nonlinear activations. However, the behavior of the full trained system cannot be predicted analytically from these operations.

Training involves optimizing billions of parameters simultaneously through stochastic gradient descent. The final configuration of those parameters is determined by training data, initialization, optimization noise, and emergent dynamics of large-scale computation. The learned circuits are not designed. They are discovered by training.

Mechanistic interpretability research has confirmed this repeatedly. Induction heads — attention heads that detect and extend repeating token sequences — were not predicted before they were observed. Name Mover heads, Subject-Tracking heads, and Predicate-Argument heads were all discovered experimentally through a combination of attention pattern inspection, behavioral probing, activation patching, and path patching across model internals.

The mathematical architecture defines the space of possible operations. It does not determine which specific algorithms the trained network will implement, nor can it predict the capabilities that will emerge at scale. This is not a gap that will be closed by more precise mathematics. It is a structural property of complex dynamical systems — the same property that makes turbulence unpredictable from the Navier-Stokes equations and cellular automata behavior unpredictable from their transition rules.

What follows is therefore not a theoretical prediction. It is an observed effect explained post hoc by known properties of transformer attention mechanisms.

3. The Intervention: A Semantic Intermediate Representation

Modern compilers do not execute raw source code directly. Source code is parsed into an intermediate representation (IR) — a structured internal form optimized for analysis, transformation, and code generation. Common IR formats include LLVM IR, Static Single Assignment form, and Abstract Syntax Trees.

The IR solves a specific problem: raw source code is ambiguous, redundant, and structured for human readability rather than machine analysis. The IR strips away these properties and exposes the essential semantic structure explicitly.

The same principle applies to natural language documents. Raw narrative text is structured for human communication. It encodes events implicitly, distributes semantic roles across variable syntactic patterns, and mixes narrative layers without boundary markers. These properties are appropriate for human readers. They impose unnecessary computational overhead on transformer inference.

A semantic IR for natural language performs the same function as a compiler IR for source code: it converts the source representation into an explicit, deterministic, machine-optimized form before the reasoning system processes it.

3.1 Structure of a Semantic Event Stream

A semantic IR for narrative documents encodes each statement as a typed event clause that exposes its semantic roles explicitly. Each clause represents a single event. The representation encodes:

- Actor: the agent performing or initiating the event
- Action: the event predicate
- Object: the entity on which the action operates
- Contextual modifiers: temporal, causal, locative, and manner relations

Example transformation:

Raw text:

The CIA retained the shellfish toxin for five years after the President ordered its destruction.

Semantic IR:

[CORPUS|00312]
d:t| +A<CIA> T<retained> O<shellfish toxin> Ct<for five years>
[CORPUS|00313]
d:t| +A<President Nixon> T<ordered> O<destruction of shellfish toxin> Ct<prior to this>

The block tag provides a stable, addressable identifier for every event. The role markers (+A, T, O, Ct) make semantic structure explicit. The representation is linear, human-readable, and directly ingestible by any transformer model without modification.

3.2 The Document-Type Classifier

A semantic IR also distinguishes between document-layer types that raw narrative conflates. Two primary types are relevant:

- Transactional blocks (d:t): events that change state — actions, decisions, transfers, orders
- Stative blocks (d:s): conditions that hold over time — roles, relationships, persistent states

This distinction matters for reasoning. A query about what happened requires access to transactional blocks. A query about who held a role at a given time requires access to stative blocks. In raw text, these are interleaved without markers. In a semantic IR, they are explicitly typed.

3.3 Entity Registry

For documents with many named entities and many surface variants, a semantic IR can include a structured entity registry that resolves coreference globally before inference begins.

The registry maps every named form and generic reference (pronoun, role descriptor, anaphora) to a canonical entity identifier. The transformer receives a flat lookup table alongside the event stream. Identity resolution — which consumes substantial attention capacity in raw text processing — is removed from the inference task entirely.

4. How the IR Changes Transformer Behavior

The semantic IR does not modify the model. It modifies the statistical structure of the input sequence. This modification interacts directly with the functional attention heads that implement transformer capabilities.

4.1 Reduced Lexical Entropy

In raw text, the same entity appears under multiple surface forms. Each variant form requires attention heads to perform identity inference. In the IR, each entity appears under its canonical form throughout. The Name Mover heads responsible for entity propagation receive a stable, low-entropy input. The inference load for identity resolution is eliminated.

4.2 Explicit Semantic Roles as Attention Anchors

Role markers in the IR (+A, T, O, C*) create predictable token positions at fixed structural locations within every clause. Predicate-Argument heads can learn stable routing patterns: predicate token attends to actor token, predicate token attends to object token, contextual modifier attends to event anchor.

In raw text, these structural positions vary with syntax. The model must solve a different parsing problem for each sentence. In the IR, the positions are invariant. Attention heads operate on a regular, repeating structure. This is precisely the regime in which transformer attention performs most reliably.

4.3 Narrative Segmentation Prevents Attention Diffusion

The document-type classifier (d:t / d:s) and the per-block addressing scheme divide the document into discrete, labeled units. Attention heads that specialize in long-range context — which preferentially attend to paragraph anchors and topic declarations — receive explicit structural landmarks at every block boundary.

Mixed-layer narrative content (events embedded in commentary embedded in background) is one of the primary causes of attention diffusion. The IR eliminates this by separating narrative layers before inference. Each block is unambiguously typed. Attention does not diffuse across heterogeneous content because heterogeneous content has been structurally separated.

4.4 Semantic Density and Effective Context Expansion

In raw text, a single complex event may require several sentences to express. In a semantic IR, the same event occupies one clause. A context window of N tokens therefore contains substantially more discrete semantic events when the input is IR-encoded than when it is raw text.

This effect — higher semantic density per token — is equivalent to expanding the effective reasoning range of the model within a fixed context window. The model does not process more tokens. It processes the same number of tokens, but each token carries more semantic information with less noise.

4.5 Externalized Working Memory

One of the most important properties of a semantic IR is that it functions as externalized working memory for the transformer. In raw text processing, the model must maintain internal representations of entities, their relationships, and the event history of the document simultaneously with active reasoning. This places a heavy load on the model's internal state.

In IR-based processing, the structure is outside the model. The transcript accumulates incrementally. The model reads what it needs from the explicit event record rather than reconstructing it from internal state. This allows the model to allocate its representational capacity primarily to inference and reasoning rather than to semantic reconstruction.

The analogy to external memory systems in software architecture is precise: just as a database offloads state management from application logic, the semantic IR offloads semantic state management from transformer inference.

5. Why the IR Can Be Generated Incrementally by the Same Models

An apparent paradox arises from the architecture described above. If transformer models struggle with long-document reasoning, how can they reliably generate a semantic IR from the same documents? The answer lies in task decomposition.

When a model is asked to reason over a long document directly, it must perform several operations simultaneously: syntactic parsing, entity resolution, long-range coreference, event extraction, and question answering. The combination overloads attention, and the failures described in Section 1 follow.

Generating a semantic IR is a fundamentally different task. The model operates one sentence at a time. Each step is a local transformation:

Input:

```
The CIA retained the toxin for five years.
```

Output:

```
+A<CIA> T<retained> O<shellfish toxin> Ct<for five years>
```

This requires only local grammatical operations: identify the subject, identify the predicate, identify the object, assign contextual modifiers. These are operations that transformers perform with high reliability. Interpretability research has established that syntactic structure is strongly represented in transformer embeddings even in models not explicitly trained for syntactic tasks (Hewitt & Manning, 2019).

The critical point is that the IR generation pipeline externalizes memory. The model never needs to hold the global document state simultaneously. The transcript accumulates outside the model. Each sentence is processed in isolation. The document-scale semantic structure emerges from the accumulation of local operations — not from any single inference step that must span the full document.

This is why only models with strong instruction-following fidelity and long-form output stability can generate reliable semantic IR at document scale. The task is not cognitively demanding in the way that interpretation is. But it demands strict structural discipline over hundreds or thousands of output clauses. Models that drift or corrupt structure over long outputs fail at the generation task for the same reason they fail at the interpretation task: loss of coherent structure over distance.

6. Theoretical Note: Semantic Attractors and Embedding Space

> **THEORY** *This section is optional. It provides a theoretical account of why IR-encoded documents are particularly well-suited to transformer vector computation.*

In pretrained transformer embedding spaces, semantically related tokens consistently form stable clusters. These clusters are not manually defined. They are not derived from ontological commitments. They emerge as a consequence of large-scale training over natural language.

Osgood's semantic differential treated meaning as position in a continuous evaluative space. Wierzbicka's Natural Semantic Metalanguage postulated a fixed inventory of universal primitives that could not be further defined — only identified through patterns of use. Transformer embedding spaces provide an empirical instantiation of a related idea: stable regions of high-dimensional space where semantically related tokens consistently cluster. We can call these regions semantic attractors.

Unlike symbolic primitives, attractors are not defined extensionally or intensionally. They are identified operationally through usage regularities across training. A semantic IR preserves and exploits these attractors by preventing premature normalization or symbolic collapse. When role-marked tokens arrive at consistent structural positions in every clause, the model's attention heads can operate in stable attractor neighborhoods rather than navigating high-entropy token sequences where attractors are obscured by lexical noise.

The IR does not define the attractors. It removes the obstacles that prevent the model from reaching them efficiently.

7. Operational Implications

The architecture described in this paper has direct consequences for any organization operating large language models over document-heavy analytical workloads.

7.1 What Improves

- Entity tracking across long documents with many actors and many surface variants
- Causal chain reconstruction over multi-step event sequences
- Temporal reasoning over documents where chronology is implicit or inverted
- Factual recall: the model can retrieve specific events by addressing the event stream rather than scanning raw narrative
- Hallucination reduction: the model is not generating bridging material to cover reconstruction failures, because reconstruction has already been completed
- Auditability: every analytical claim can be traced to a specific addressed clause in the transcript

7.2 What Does Not Change

A semantic IR does not expand the nominal context window. It does not modify model weights. It does not add symbolic inference capabilities. It does not resolve ambiguity or infer intent. It prepares text so that the model's existing capabilities can operate over explicit structure rather than implicit structure.

Inference, synthesis, judgment, and question answering remain the model's responsibility. The IR removes the overhead of semantic reconstruction from the inference task. It does not perform inference itself.

7.3 Applicable Domains

The architecture is most valuable where documents are long, entities are numerous, events are densely interconnected, and analytical errors carry material consequences. This includes:

- Intelligence analysis: HUMINT reports, OSINT corpora, multi-source dossier construction
- Legal and compliance reasoning: contract analysis, regulatory review, investigative records
- Defense and operations: after-action reports, mission debriefs, incident timelines
- Scientific and technical review: research corpora, standards documents, patent analysis

In each of these domains, the properties that make raw narrative difficult for transformers — lexical variability, implicit structure, heterogeneous discourse layers, dense entity populations — are precisely the properties that structured semantic IR is designed to eliminate.

8. Conclusion

The effective reasoning range of large language models over long documents is substantially shorter than their nominal context window. This is not a scale problem. It is a structural problem: transformers must simultaneously reconstruct semantic structure and reason over it, using the same attention mechanism for both tasks. The entanglement degrades with distance.

A semantic intermediate representation separates these tasks. Semantic structure is compiled before inference begins. The model receives an explicit event stream with stable entity references, typed clause roles, and addressable block identifiers. Its attention heads operate on a regular, low-entropy input that is structurally optimized for the operations they already perform.

The result is a measurable increase in effective reasoning range within an unchanged context window, and a corresponding improvement in entity tracking, causal reasoning, factual recall, and hallucination resistance.

This is not a claim about a new model capability. It is a claim about input representation. The same model, operating on a better-structured input, performs substantially better analytical work. The principle is as old as compilers.

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